Assignment 2　杜奕旋　122090095

1. suppose that head: 1　tail: 0

(a) $\hat{p} = \frac{159}{314}$　s.e. of $\hat{p} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{\frac{159}{314} \times \frac{155}{314}}{314}}$

The 90% CI: $0.5063 \pm 1.645 \sqrt{\frac{0.5063 \times 0.4937}{314}} = [0.4599, 0.5528]$

(b) $n = \left(\frac{1.645}{0.01}\right)^2 \cdot 0.5064 \cdot (1-0.5064) \approx 6764$

(c) $Z_{\frac{\alpha}{2}} \sqrt{\frac{p(1-p)}{n}} = \frac{0.11}{2}$　$Z_{\frac{\alpha}{2}} = 1.9494$

$\therefore \alpha = 0.05$　$\therefore$ 95% CI.

2. (a) $100(1-\alpha)\%$ CI for $\mu_x - \mu_y$: $\left[\bar{x}-\bar{y}-Z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}, \bar{x}-\bar{y}+Z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}\right]$

(b) Interval $= 2Z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}$

$\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m} = \frac{2500}{n}+\frac{900}{6000-n}$　16n=60000　$\therefore n = \frac{15000}{4} = 3750$　$m = 2250$

3. (a) 90% CI: $41.75 \pm 1.645\left(\frac{11}{\sqrt{12}}\right) = [36.5264, 46.9736]$

(b) 95% CI: $41.75 \pm 1.96\left(\frac{11}{\sqrt{12}}\right) = [35.5262, 47.9738]$

99% CI: $41.75 \pm 2.576\left(\frac{11}{\sqrt{12}}\right) = [33.5701, 49.9299]$

(c) X is normally distributed, σ is not known: $\bar{X} \pm t_{\frac{\alpha}{2}(n-1)}\left(\frac{s}{\sqrt{n}}\right)$

$\therefore$ 90% CI: $41.75 \pm t_{0.05}(11)\left(\frac{11.8}{\sqrt{12}}\right)$

$= [35.6322, 47.8678]$

4 (a) $\because \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} \sim \chi^2(n)$ $\quad P\left(\chi^2_{1-\frac{\alpha}{2}}(n) < \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} < \chi^2_{\frac{\alpha}{2}}(n)\right) = 1-\alpha$

$\Leftrightarrow \frac{1}{\chi^2_{\frac{\alpha}{2}}(n)} < \frac{\sigma^2}{\sum_{i=1}^{n}(X_i-\mu)^2} < \frac{1}{\chi^2_{1-\frac{\alpha}{2}}(n)}$

$\therefore \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{\frac{\alpha}{2}}(n)} < \sigma^2 < \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{1-\frac{\alpha}{2}}(n)}$ $\quad \therefore CI = \left[\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{\frac{\alpha}{2}}(n)}, \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{1-\frac{\alpha}{2}}(n)}\right]$

(b) $\because \sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$

$P\left(\chi^2_{1-\frac{\alpha}{2}}(n-1) < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\frac{\alpha}{2}}(n-1)\right) = 1-\alpha \Leftrightarrow \frac{1}{\chi^2_{\frac{\alpha}{2}}(n-1)} < \frac{\sigma^2}{(n-1)S^2} < \frac{1}{\chi^2_{1-\frac{\alpha}{2}}(n-1)}$

$\therefore \frac{1}{\chi^2_{\frac{\alpha}{2}}(n-1)} < \frac{\sigma^2}{\sum(X_i-\bar{X})^2} < \frac{1}{\chi^2_{1-\frac{\alpha}{2}}(n-1)}$

$\therefore \frac{\sum(X_i-\bar{X})^2}{\chi^2_{\frac{\alpha}{2}}(n-1)} < \sigma^2 < \frac{\sum(X_i-\bar{X})^2}{\chi^2_{1-\frac{\alpha}{2}}(n-1)}$ $\quad \therefore CI = \left[\frac{\sum(X_i-\bar{X})^2}{\chi^2_{\frac{\alpha}{2}}(n-1)}, \frac{\sum(X_i-\bar{X})^2}{\chi^2_{1-\frac{\alpha}{2}}(n-1)}\right]$

(c) $CI = \left[\sqrt{\frac{S^2}{\chi^2_{\frac{\alpha}{2}}(n-1)}}, \sqrt{\frac{S^2}{\chi^2_{1-\frac{\alpha}{2}}(n-1)}}\right]$ when μ is unknown

5 (a) $\bar{X} = \frac{21+22+18+29+35+20+30+32+26+23+19+29+31+28+27}{15} = 26$

$S^2 = \frac{1}{14}\sum_{i=1}^{15}(X_i-\bar{X})^2 = 27.14$

(b) $H_0: \mu = \mu_0 = 26$

$H_1: \mu > \mu_0$

(c) $Z = \frac{\bar{X}-\mu_0}{\frac{\sigma}{\sqrt{n}}} = 0$ $\quad Z_{0.05} \approx 1.645$

(d) Charles would not reject H_0

(e) $\frac{\bar{X}-26}{\frac{4}{\sqrt{15}}} > 1.645$ $\quad \bar{X} > 27.6989$

(f) when σ^2 is not known, $t = \frac{\bar{X}-\mu_0}{\frac{S}{\sqrt{n}}} = 0$

Charles would not reject H_0.

6. (a)

```python
import pandas as pd

# Load the dataset
data = pd.read_csv("traffic.csv")

# Filter data for "Clear" and "Rain" weather conditions
clear_traffic = data[data['weather_main'] == "Clear"]
rain_traffic = data[data['weather_main'] == "Rain"]

# Calculate sample means
x_mean = clear_traffic['traffic_volume'].mean()
y_mean = rain_traffic['traffic_volume'].mean()

print("Sample mean of traffic volume when Clear:", x_mean)
print("Sample mean of traffic volume when Rain:", y_mean)

```

$\therefore \bar{x} = 3055.9088$

$\bar{y} = 3317.9055$

```
PS C:\Users\HUAWEI\Desktop\sta2002> & C:/Users/HUAWEI/AppData/
002/sta_code2_122090095.py
Sample mean of traffic volume when Clear: 3055.908819356284
Sample mean of traffic volume when Rain: 3317.9055007052184
PS C:\Users\HUAWEI\Desktop\sta2002>
```

(b) $\bar{x} - \bar{y} \pm z_{0.025}(\text{pooled std})$

$$\text{pooled std} = \sqrt{\frac{(n-1)s_x^2 + (m-1)s_y^2}{n+m-2}\left(\frac{1}{n}+\frac{1}{m}\right)}$$

```
(b) 均值差: -261.99668134893454
(b) 95% 置信区间: (-323.65226908136776, -200.34109361650133)
```

(c) $\bar{x} - \bar{y} \pm z_{0.025}\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}$

```
95% Welch's t-Interval for the difference μX - μY:
Lower Bound: -323.5949128379981
Upper Bound: -200.39844985987102
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7. (a) $H_0: \mu_Y \geq \mu_X$

$H_1: \mu_Y < \mu_X$

(b) Assume $\sigma_X^2 = \sigma_Y^2$, $t = \frac{(\bar{x}-\bar{y})}{\sqrt{\frac{s_p^2}{n} + \frac{s_p^2}{m}}}$ $\quad s_p^2 = \frac{(n-1)s_x^2 + (m-1)s_y^2}{n+m-2}$

(c) Assume $\sigma_X^2 \neq \sigma_Y^2$, $t = \frac{(\bar{x}-\bar{y})}{\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}}$

(d) $H_0: \sigma_X^2 = \sigma_Y^2$

$H_1: \sigma_X^2 \neq \sigma_Y^2$ $\quad F = \frac{s_x^2}{s_y^2}$

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(b) t_statistic is:-8.336873582326676  t_critical is:-1.644933572581743
 So reject H0: μX > μY
(c) t_statistic_welch is:-8.336873582326676  t_critical_welch is:-1.6449959837678225
So reject H0: μX > μY
(d) f_statistic is:1.0049224336665912 f_critical_low is:0.957221125506382 f_critical_high is:1.0451043846030437
Fail to reject H0
```